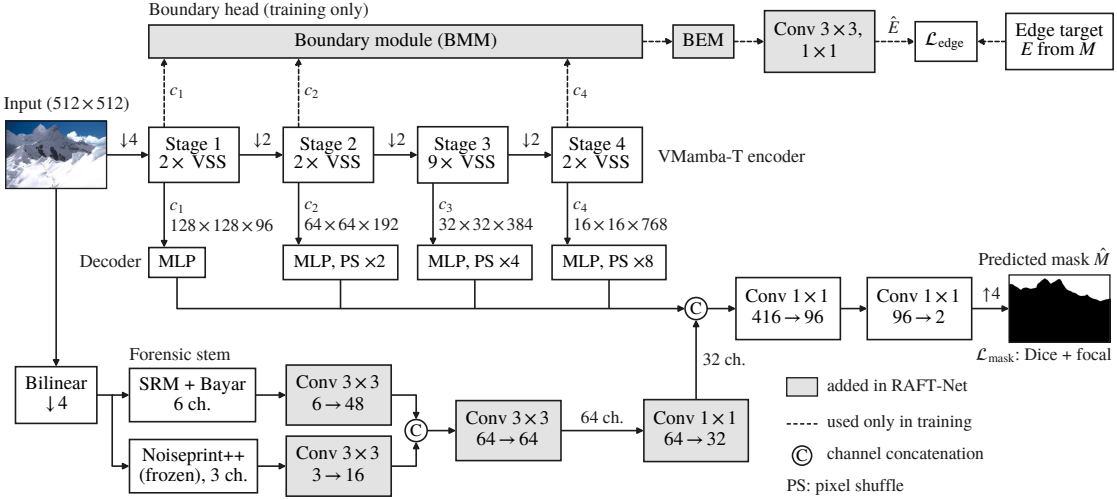

Boundary head (training only)
Boundary module (BMM)
BEM
Conv 3×3, 1×1
$\hat{E}$
$\mathcal{L}_{\text{edge}}$
Edge target E from M
Input (512×512)
c_1
c_2
c_4
↓4
Stage 1 2× VSS
↓2
Stage 2 2× VSS
↓2
Stage 3 9× VSS
↓2
Stage 4 2× VSS
VMamba-T encoder
c_1 128×128×96
c_2 64×64×192
c_3 32×32×384
c_4 16×16×768
Decoder
MLP
MLP, PS ×2
MLP, PS ×4
MLP, PS ×8
Predicted mask $\hat{M}$
Conv 1×1 416→96
Conv 1×1 96→2
↑4
$\mathcal{L}_{\text{mask}}$: Dice + focal
32 ch.
Bilinear ↓4
Forensic stem
SRM + Bayar 6 ch.
Noiseprint++ (frozen), 3 ch.
Conv 3×3 6→48
Conv 3×3 3→16
Conv 3×3 64→64
64 ch.
Conv 1×1 64→32
added in RAFT-Net
used only in training
channel concatenation
PS: pixel shuffle